

HW2

題目一：對 `cycleData_peak.txt`：Count how many not “NA” in each row.

```
data <- read.delim("cycleData_peak.txt") # 讀檔
dim(data) # 觀察維度
```

Output:

```
[1] 800 79
```

撰寫函式搭配 `apply`

```
count_NA <- function(x){
  length(x) - sum(is.na(x))
}

row_Non_NA_nums <- apply(data, 1, count_NA)
row_Non_NA_nums
```

Output:

```
[1] 72 77 78 76 77 75 75 72 76 71 78 73 72 78 76 74 78 79 79 79 78 79 78 79 77 54 56 77 78 77 78 78 78 78 78 77 75 78 53 56 54 77 73 78 79 78 77 78
[51] 78 78 78 70 76 78 78 76 77 77 77 71 79 77 75 78 78 79 79 78 45 79 77 78 78 56 78 78 78 70 77 79 79 77 74 79 79 58 77 77 78 79 75 78 77 78 79 78 78 74
[101] 78 77 77 77 78 78 78 77 78 78 78 77 79 79 78 78 74 79 78 78 77 66 79 74 54 78 74 78 77 78 78 76 77 77 78 79 79 77 77 79 76 77 79 76 77 78 78
[151] 77 77 77 78 78 76 78 76 74 75 77 78 78 77 78 78 54 77 78 78 76 78 77 75 77 77 79 78 76 78 78 78 76 78 77 78 78 76 77 78 78 78 78 77 76
[201] 78 76 77 77 76 78 76 78 77 78 77 44 77 77 78 76 77 77 45 73 74 75 78 76 66 62 78 56 77 78 76 78 56 79 76 78 76 78 76 78 78 77 77 68 75 76 59 78 77 78
[251] 78 77 79 78 75 78 70 78 74 77 76 78 77 78 77 78 79 78 78 76 77 75 78 78 73 78 78 78 77 69 78 72 78 76 78 68 77 71 76 77 79 78 78 78 76 66 77 78 77 78
[301] 78 78 77 71 76 77 73 71 78 78 74 77 77 56 78 68 53 78 78 78 78 79 78 68 77 78 77 77 78 53 76 78 78 78 78 77 78 77 78 78 57 77 77 79 79 78 78 78
[351] 78 77 78 78 79 78 79 78 78 77 78 77 77 78 78 78 76 74 77 79 78 78 78 79 77 79 76 76 79 79 55 78 68 78 78 78 77 77 79 77 78 77 79 78 77 78 78 78 78
[401] 79 78 78 75 77 78 78 78 78 77 77 77 77 78 78 56 78 78 77 77 78 77 78 79 19 77 44 78 79 78 78 78 78 78 77 77 78 78 76 78 77 25 78 78 77 78 78
[451] 78 78 79 78 55 77 78 77 69 77 79 78 77 79 77 66 77 74 78 78 78 75 78 79 75 77 78 78 75 76 78 77 78 78 76 69 64 78 77 76 76 77 78 77 75 78 72 53 74 76
[501] 78 77 78 78 78 79 78 78 78 78 75 56 78 78 77 78 78 77 78 77 76 77 79 65 79 53 56 68 75 79 57 78 79 78 77 78 78 76 75 78 78 62 68 78 76 79 78 78
[551] 77 78 78 79 78 78 77 78 78 77 69 78 78 64 63 78 78 78 78 77 77 76 76 73 78 77 78 78 58 78 78 76 56 77 72 79 77 77 78 77 78 72 77 74 77 78 78 76
[601] 79 78 70 79 75 67 76 78 76 77 78 77 76 74 77 78 78 45 77 77 75 76 76 77 73 78 78 78 57 78 79 78 76 78 78 79 78 44 78 79 78 77 78 78 78 78 77 78 78
[651] 76 77 78 78 77 78 77 78 75 78 59 78 78 76 76 58 75 77 78 75 78 78 77 74 77 76 78 77 77 75 78 78 79 77 78 78 77 78 78 78 77 79 77 76 54 78 76 78 79
[701] 76 78 78 78 78 78 79 79 77 78 77 78 57 53 79 75 78 78 77 76 77 77 76 78 77 77 78 78 78 75 78 78 78 78 78 78 76 76 78 78 78 78 77 78 77
[751] 78 78 78 75 77 78 78 78 78 78 78 77 79 77 78 78 78 77 78 77 77 78 76 66 78 78 75 77 78 70 78 75 78 77 76 78 78 78 78 76 71 78 79 79 78 75
```

題目二：對 SRBCT_train.txt：

2-1 Normalize data by each row (let values b/w 0~1，min-max)

```
data2 <- read.delim("SRBCT_train.txt")
data2_without_ch <- as.matrix(data2[,-c(1,2)])
dim2 <- dim(data2_without_ch)
dim2
```

Ouput:

```
[1] 2308    63
```

```
## min_max
min_max_norm <- function(x){
  (x - min(x)) / (max(x) - min(x))
}

## 使用 apply
min_max_each_row <- t(apply(data2_without_ch, 1,
                             min_max_norm))
min_max_each_row
```

Ouput:

```
      EWS.T1      EWS.T2      EWS.T3      EWS.T4      EWS.T6      EWS.T7      EWS.T9      EWS.T11      EWS.T12      EWS.T13      EWS.T14      EWS.T15
[1,] 0.883321121 0.447099938 0.9045713319 0.2642748436 0.744461981 0.560988670 0.501662815 0.74564568 0.644608534 0.5244349248 1.0000000000 0.5869736768
[2,] 0.009971738 0.011518157 0.0355143177 0.0752946195 0.099877353 0.017543860 0.039460353 0.069900881 0.015890791 0.1080360476 0.0299685384 0.0251693062
[3,] 0.292887029 0.291206800 0.2423483675 0.0900075775 0.069943663 0.245478206 0.277171943 0.11181761 0.199123645 0.1375481830 0.1404474022 0.1127400916
[4,] 0.041532629 0.000000000 0.0283777574 0.0295840993 0.095932904 0.044117647 0.096679688 0.11425781 0.077435662 0.0480238971 0.1307444853 0.0971966912
[5,] 0.624226592 0.226041523 0.3323250378 0.3551491819 0.455107933 0.489481644 0.462670150 0.59851506 0.517805582 0.4072597278 0.5569916128 0.5776158394
[6,] 0.247389999 0.162400050 0.1461423347 0.1198519450 0.128671490 0.123349656 0.126466604 0.25891916 0.220931365 0.2007066262 0.2402351879 0.2398101496
[7,] 0.100110025 0.017727749 0.0707091751 0.2420036328 0.069315955 0.000000000 0.080416143 0.29764780 0.182140741 0.3410003841 0.1338858471 0.2789369861
[8,] 0.670018638 0.369078063 0.4832040224 0.000000000 0.165185731 0.548350743 0.822417754 0.19700056 0.162281652 0.4593645702 1.0000000000 0.2276450956
[9,] 0.245725779 0.070197787 0.1431780087 0.0705330204 0.109587663 0.043714381 0.176366074 0.07448877 0.179249078 0.4154877640 0.1577941669 0.1860207844
[10,] 0.939823380 0.625257426 0.6526929387 0.5073126462 0.165171559 0.452441621 0.488428915 0.29215679 0.273098537 0.4234353730 0.6263743935 0.2345980663
[11,] 0.125397994 0.000000000 0.1951877112 0.2037551107 0.168270467 0.366039041 0.456555480 0.38428246 0.357436166 0.0986403909 0.4428973065 0.3624914636
[12,] 0.592988006 0.454461579 0.1721365494 0.9371952023 0.317385001 0.243640438 0.275998418 0.13384737 0.376103862 0.5871227099 0.3465137736 0.1486094636
[13,] 0.328985938 0.394672212 0.3930481283 0.8652406417 0.287215290 0.365290156 0.294434541 0.46951872 0.407833234 0.2142998614 0.5549316696 0.8286987522
[14,] 0.323485711 0.572458026 0.6409043633 0.000000000 0.403397553 0.480313011 0.541776039 0.51357040 0.428086771 0.5738943093 0.5900153534 0.4571343668
[15,] 1.000000000 0.485935246 0.6485815034 0.3341080301 0.296481808 0.367326495 0.378065395 0.39842122 0.403670460 0.2944782818 0.4111235775 0.3862397820

      EWS.T19      EWS.C8      EWS.C3      EWS.C2      EWS.C4      EWS.C6      EWS.C9      EWS.C7      EWS.C1      EWS.C11      EWS.C10
[1,] 0.632433465 2.821431e-01 0.2397553689 2.413900e-01 0.7209571050 0.2847359224 0.290400767 0.4266670424 0.5914830055 0.0875091596 2.948255e-01
[2,] 0.0589239055 7.876073e-02 0.0760944915 2.127660e-02 0.0229829894 0.0683090705 0.062230043 0.0037327361 0.0202100997 0.0246360582 4.991201e-02
[3,] 0.1388989556 5.022238e-01 0.2070964979 4.460844e-01 0.5639640233 0.7211148816 0.613547261 0.3256679735 0.1516159852 0.1708892037 5.486443e-01
[4,] 0.2609145221 6.991039e-01 0.0025275735 4.193474e-03 0.0708295037 0.7246668199 0.676011029 0.8317440257 0.0079273897 0.4689223346 5.139697e-01
[5,] 0.5710160869 2.847518e-01 0.5098308813 6.799120e-01 0.5352674275 0.1629313901 0.157156607 0.1161813431 0.3519868005 0.1669187405 1.343325e-01
[6,] 1.0000000000 7.278781e-02 0.0572030709 6.456155e-02 0.1999450992 0.0939688837 0.089178348 0.0726904039 0.1736812744 0.0364204691 7.081315e-02
[7,] 0.5584403747 2.624788e-01 0.0660672782 1.304484e-01 0.0529098118 0.0860476169 0.124152838 0.2270623238 0.1078183085 0.1498883471 2.722183e-01
[8,] 0.4323609726 1.911491e-01 0.3120801006 2.561224e-01 0.8556629535 0.8092843830 0.798925057 0.5161024663 0.3306748732 0.2243942612 4.214382e-01
[9,] 0.0257458934 1.410606e-01 0.2292658398 3.524304e-01 0.0752933289 0.1688568555 0.145491116 0.1740529668 0.2062353336 0.2105598391 1.418706e-01
[10,] 0.2699302985 2.830814e-02 0.5010646096 3.367657e-01 0.8713742190 0.3885999511 0.327480889 0.4238891410 0.5419386366 0.0000000000 1.986457e-01
[11,] 0.3003822515 3.215968e-01 0.1238725355 4.720052e-02 0.0129309198 0.0379590787 0.093682651 0.2510709249 0.1656541289 0.1997463482 2.715050e-01
[12,] 0.3299723211 6.293606e-02 0.2792276262 8.559378e-01 0.6072887834 0.3361671280 0.324831949 0.3433504679 0.8684592065 0.166607645 4.303414e-02
[13,] 0.4548916617 2.784314e-01 0.4242820360 3.960487e-01 0.2087344029 0.2508912656 0.324341454 0.5410576352 0.4297979798 0.3516141810 2.326896e-01
[14,] 1.0000000000 1.363974e-01 0.1598484473 2.083849e-01 0.6132930514 0.3348769254 0.292506562 0.3083552078 0.5199593878 0.1871625972 1.991977e-01
[15,] 0.4260298125 2.01038e-01 0.3145135438 3.503767e-01 0.2185446386 0.1529892611 0.155794198 0.1501843244 0.2857028370 0.2217903510 1.944222e-01

      BL.C5      BL.C6      BL.C7      BL.C8      BL.C1      BL.C2      BL.C3      BL.C4      NB.C1      NB.C2      NB.C3
[1,] 0.0649906995 0.0330590158 0.0102023561 0.0703173440 0.0212502114 0.0791950848 0.0877346260 0.000000000 0.3333521222 0.330843808 0.2017924581
[2,] 0.0183970565 0.0420732683 0.0266624007 0.0000000000 0.0036794113 0.0033594625 0.0077854210 0.037807284 0.0932117528 0.041753319 0.0495387405
[3,] 0.3103152901 0.5272654441 0.0260929727 0.2682107205 0.3020788719 0.5736171054 0.4082298290 0.143511350 0.4331367575 0.900405232 0.3313346292
[4,] 0.7252412684 0.7658359375 0.6953699449 0.6988166360 0.0389476103 0.0415326287 0.1530905331 0.035443474 0.0278607537 0.039035055 0.0545151654
[5,] 0.3965351299 0.1214079472 0.1768183693 0.0236491132 0.4859067785 0.4084971814 0.5800907466 0.822494156 0.5432421284 0.6132227004 0.3541867180
[6,] 0.0377310039 0.0182058071 0.0569285670 0.0634281110 0.1782238712 0.1485951599 0.1582913460 0.123464771 0.2230388467 0.0949606640 0.1973948694
[7,] 0.0352926088 0.1717697150 0.1599989583 0.1018613160 0.075528935 0.1531370238 0.0767898646 0.106854773 0.0875775448 0.050481442 0.2299789715
[8,] 0.8376750033 0.3690347189 0.1598977071 0.1242685623 0.3462788782 0.8340340687 0.5808590872 0.398162195 0.4850244896 0.239868233 0.0644965541
[9,] 0.3140797855 0.7686892390 0.5655715722 0.2256788468 0.5834394904 0.3086490111 0.3767348307 0.433489775 0.0000000000 0.591015756 0.3985920215
[10,] 0.6409298754 0.8516876680 0.4360361618 0.5050089008 0.5823938008 0.9032426961 1.0000000000 0.587769207 0.5903870990 0.441306852 0.2485950644
[11,] 0.1579292790 0.1695032505 0.2848527312 0.2721524039 0.1821237572 0.2304417621 0.1274644577 0.176917687 0.1261429851 0.093735865 0.4086543152
[12,] 0.2732964281 0.4062870700 0.4331092658 0.4644128114 0.8001845262 0.6631079478 0.6437327007 0.770001318 0.6244233557 0.446553315 0.5655726901
[13,] 0.3044919786 0.4411665676 0.3871302436 0.5399683106 0.4073578927 0.4556149733 0.3505743712 0.448009507 0.3348286789 0.355684794 0.7293325411
[14,] 0.4319251152 0.3328463177 0.2707642019 0.0873408945 0.1507107127 0.2513496112 0.4333347432 0.217374078 0.2104898222 0.217671240 0.1720568570
[15,] 0.1229764385 0.0943660843 0.0690415131 0.1117566918 0.2789309184 0.4522760058 0.6525084148 0.920339798 0.5218384356 0.576494631 0.2365363039
```

```

      NB.C6      NB.C12      NB.C7      NB.C4      NB.C5      NB.C10      NB.C11      NB.C9      NB.C8      RMS.C4      RMS.C3
[1,] 0.1838960600 0.4613888732 0.3598726115 0.1660560284 0.1472859478 0.3263626628 0.3399188321 2.458711e-01 1.373091e-01 0.7672340905 0.570063694
[2,] 0.0172239108 0.0892123927 0.0412733963 0.0151975684 0.0410600971 0.0427664907 0.0042126593 4.185997e-02 7.732099e-02 0.0116781315 0.159014558
[3,] 0.1587651962 0.4578130663 1.0000000000 0.4085263401 0.0542615227 0.3694857181 0.3757124502 5.604059e-01 1.812341e-01 0.2616215860 0.003689915
[4,] 0.8998736213 1.0000000000 0.8588005515 0.4964958640 0.6954848346 0.6142578125 0.7945197610 7.517233e-01 9.093520e-01 0.1745174632 0.086282169
[5,] 0.1082084422 0.2094046473 0.0856592878 0.0747971951 0.1348824419 0.1066959989 0.0375360924 1.542692e-01 1.963426e-01 0.1307575966 0.447545717
[6,] 0.0865218585 0.0584604759 0.0307355819 0.0465550548 0.0747359007 0.1249169847 0.1081191170 8.959453e-02 1.205603e-01 0.4240111218 0.261540233
[7,] 0.2114048737 0.2280128385 0.1271931823 0.1750053711 0.1552724266 0.1249210617 0.3086047617 1.682932e-01 2.747378e-01 0.1532932728 0.064967025
[8,] 0.1467209917 0.3648303064 0.1809197694 0.1279961857 0.1163365264 0.3686446188 0.1621082745 1.460708e-01 1.732911e-01 0.7276667678 0.476659009
[9,] 0.1738853503 0.2984244050 0.7296345960 0.0119342943 0.0770700637 1.0000000000 0.7818303721 4.287965e-01 7.355012e-02 0.0972510895 0.130271539
[10,] 0.4231212259 0.6969876785 0.5605431254 0.2815106985 0.2985444518 0.3873433628 0.3201856958 5.236832e-01 3.393836e-01 0.6795350623 0.201682432
[11,] 0.2168900162 0.2401798622 0.1827268454 0.1070126737 0.1416192917 0.1380805832 0.2902184421 2.607913e-01 3.746330e-01 0.3389887630 0.259567373
[12,] 0.3420983261 0.3125741400 0.3136944774 0.4825359167 0.5629366021 0.5216818242 0.8154079346 5.356531e-01 5.811915e-01 0.2806115724 0.323711612
[13,] 0.4519706873 0.3885224797 0.3682115270 0.3243216479 0.3576945930 0.3539413745 0.5469597940 5.093187e-01 5.005150e-01 0.7140225787 0.263220440
[14,] 0.1781982071 0.3571145560 0.5439799911 0.2889406171 0.3688276955 0.3224951711 0.2874052796 3.113268e-01 3.770987e-01 0.6084641672 0.238101134
[15,] 0.0590639526 0.1161243789 0.1366404873 0.0991745472 0.0415932040 0.1289068761 0.1200512903 7.641449e-02 1.775525e-01 0.2031976278 0.307140567

      RMS.C9      RMS.C2      RMS.C5      RMS.C6      RMS.C7      RMS.C8      RMS.C10      RMS.C11      RMS.T1      RMS.T4      RMS.T2      RMS.T
[1,] 3.637619e-01 0.3663547714 0.330449242 0.2204498055 0.389183248 0.253762471 0.4150555211 0.669071642 0.2948537287 0.956907728 0.567414464 0.85479961
[2,] 1.250467e-01 0.0620167440 0.153735402 0.1648269610 0.060630299 0.466378713 0.0094918146 0.521409908 0.7106063030 0.659147870 0.460886258 0.18215752
[3,] 0.000000e+00 0.1883174645 0.018449577 0.2881099068 0.075280862 0.133924159 0.1686488980 0.257733997 0.0346917932 0.058808026 0.139537869 0.07798240
[4,] 8.785616e-01 0.7899816176 0.729032629 0.7423598346 0.613166360 0.014705882 0.6248276654 0.032743566 0.0122357537 0.021426930 0.008789063 0.13700597
[5,] 1.944177e-01 0.1088959164 0.241853430 0.1663687612 0.160593978 1.000000000 0.000000000 0.462395160 0.7569091159 0.387460470 0.887529218 0.09789632
[6,] 7.795025e-02 0.0903472032 0.039687951 0.0543694823 0.053634520 0.094633006 0.0219691670 0.186042805 0.1453808077 0.182270590 0.136534698 0.47551159
[7,] 2.386899e-01 0.2979928516 0.205897097 0.1299405603 0.307771434 0.051874662 0.2562353110 0.089341866 0.1391657606 0.126184074 0.121040879 0.21529807
[8,] 2.051493e-01 0.3859824021 0.249577392 0.4735815526 0.172684322 0.696285380 0.0999523211 0.815179229 0.2141649690 0.733344892 0.422565125 0.53842486
[9,] 6.466644e-02 0.2107274556 0.141434797 0.1691250419 0.247234328 0.347267851 0.0616828696 0.364163594 0.2795172645 0.129433456 0.300905129 0.07888032
[10,] 8.722818e-02 0.2330273308 0.298823694 0.7760480296 0.289922859 0.955495829 0.2885615554 0.973855981 0.4653914622 0.922684910 0.693427345 0.37526615
[11,] 1.000000e+00 0.5771465061 0.095580605 0.0821618937 0.321649978 0.037010102 0.4433407537 0.131881192 0.1984426135 0.138062845 0.148253262 0.17100210
[12,] 0.000000e+00 0.1991564518 0.334717280 0.3881639647 0.531633057 1.000000000 0.3344536708 0.740147621 0.9716620535 0.656649532 0.738500066 0.34671148
[13,] 1.361854e-01 0.2052188552 0.415517924 0.2434442464 0.404822737 0.325757576 0.6645276292 0.326381462 0.4977322242 0.491731036 0.395563478 0.68612596
[14,] 4.387846e-01 0.4028032292 0.299118419 0.4178594423 0.283765044 0.269278391 0.2363924521 0.804293992 0.1623000347 0.693031549 0.571442722 0.48846020
[15,] 1.152028e-01 0.1035422343 0.106467383 0.0439974355 0.185045680 0.550328578 0.000000000 0.436408078 0.8093444462 0.575773361 0.769634557 0.42743228

      RMS.T7      RMS.T8      RMS.T5      RMS.T3      RMS.T10      RMS.T11
[1,] 5.520827e-01 0.609604870 0.5047911617 0.338819683 0.3405388648 0.5528155121
[2,] 2.670506e-01 1.000000000 0.2530795073 0.221991148 0.3879912547 0.3505039194
[3,] 6.977893e-02 0.045234408 0.1725694330 0.256119659 0.0211511218 0.2921951702
[4,] 1.784812e-01 0.071346507 0.1432100184 0.009191176 0.0523897059 0.0284926471
[5,] 7.881204e-01 0.474907191 0.4963563866 0.485219304 0.1743434621 0.2400659975
[6,] 4.858719e-01 0.449203496 0.2830312315 0.259981759 0.0130610727 0.0000000000
[7,] 3.161503e-01 0.323363780 0.3326866362 0.171743674 0.4764682522 1.0000000000
[8,] 1.507954e-01 0.411815699 0.2629274847 0.533440250 0.6417147068 0.8434831607
[9,] 1.005699e-03 0.262655045 0.4670130741 0.351223600 0.2203821656 0.4052966812
[10,] 7.241090e-01 0.545673496 0.2224859506 0.486229886 0.3788613913 0.3943593145
[11,] 8.630812e-01 0.351174692 0.1146399652 0.317419492 0.5722065045 0.6739421567
[12,] 6.727297e-01 0.652563596 0.3519836563 0.349018057 0.4934097799 0.0561486754
[13,] 0.000000e+00 0.484600911 0.4930184195 0.413982967 0.7263913646 1.0000000000
[14,] 4.415086e-01 0.814868010 0.3917339409 0.784854638 0.6364716953 0.7182655639
[15,] 2.687931e-01 0.534300369 0.3282176631 0.823208848 0.1823609553 0.0477640648
[ reached 'max' / getOption("max.print") -- omitted 2293 rows ]

```

2-2 Normalize data by each column (mean = 0, sd = 1)

```

# z-score
z_norm <- function(x){
  (x - mean(x)) / sd(x)
}

## 使用 apply
z_score_each_col <- apply(data2_without_ch, 2, z_norm)
z_score_each_col

```


